

Rupak Kumar

☎ +91 6207861650 ✉ rupakkumar76319@gmail.com
🌐 LinkedIn 🐙 GitHub 📁 Portfolio

Summary

Builds machine learning systems end to end — from raw data cleaning to model deployment. Compared six deep learning architectures for ASL gesture recognition, with 1D-CNN reaching 92.68% accuracy on a 36-class dataset. Analyzed 541K+ transaction records using SQL and Power BI to surface business patterns. Deployed a rainfall prediction system through Flask with a modular, reusable pipeline.

Technical Skills

Programming: Python, SQL, C++

Deep Learning: CNN, LSTM, GRU, BiLSTM, CNN-BiLSTM, CNN-GRU

Frameworks & Libraries: PyTorch, Scikit-learn, OpenCV, MediaPipe, Flask, Pandas, NumPy, Matplotlib, Jupyter

Machine Learning: Regression, Classification, Model Evaluation, Feature Engineering, Exploratory Data Analysis

Data Visualization: Power BI, DAX

Tools: Git, GitHub, VS Code

Education

B.Tech in Electronics and Communication Engineering

2023 – 2027

Assam University, Silchar

- Current GPA: **7.53 / 10**

BS in Data Science

2024 – 2028

Indian Institute of Technology, Madras

Relevant Coursework

Machine Learning, Deep Learning, Probability and Statistics, Data Structures and Algorithms, Database Management Systems

Projects

Sequence-Based Dynamic Hand Gesture Recognition using Deep Learning 🐙 *GitHub* *Python, PyTorch, OpenCV, MediaPipe*

- Built a sequence-based ASL gesture recognition system using MediaPipe for 3D hand keypoint extraction.
- Trained and evaluated six deep learning models including LSTM, GRU, BiLSTM, CNN-GRU, and CNN-BiLSTM on a 36-class dataset.
- 1D-CNN achieved 92.68% test accuracy, while CNN-BiLSTM achieved lower inference time for real-time prediction.
- Built a real-time inference pipeline using OpenCV with confidence thresholding and temporal smoothing.

Weather Prediction ML System

Python, Scikit-learn, Flask

- Developed an **end-to-end ML pipeline** for rainfall prediction using Python and Scikit-learn, achieving **~84.7% accuracy**.
- Identified humidity and atmospheric pressure as strongest predictors through correlation analysis and feature importance.
- Compared Logistic Regression, Random Forest, and XGBoost; selected best model based on accuracy, precision, and recall.
- Integrated the trained model with a **Flask API** to enable real-time prediction and user interaction.

E-Commerce Sales Performance & Customer Analytics

Power BI, SQL

- Analyzed **541K+ transaction records** using SQL and Power BI to uncover sales trends and customer behavior patterns.
- Identified that the **top 20% of customers contributed ~60–70% of total revenue**, enabling targeted marketing and retention strategies.
- Developed interactive **KPI dashboards** with DAX measures (MoM, YoY growth, AOV), improving business decision-making efficiency.
- Performed **customer segmentation** to classify high-value users, supporting personalized marketing and revenue optimization.

Certifications

OpenCV Bootcamp — OpenCV University

Grade: 100%

Issued by Satya Mallick (CEO, OpenCV.org) & Gary Bradski (President, OpenCV.org)

SQL (Intermediate) Certificate | *HackerRank* | 📄 View Certificate2025